

A two-sided quasihomogeneous zero-product theorem

Candidate partial result for arXiv:0712.0167

13 August 2026

Status. Candidate substantial partial result, likely valid; human review is needed. The unrestricted Bergman zero-product problem remains open.

1 Source problem and exact literature boundary

For a bounded domain $\Omega \subset \mathbb{C}^n$, let $P : L^2(\Omega) \rightarrow A^2(\Omega)$ be the Bergman projection and $T_f h = P(fh)$. Le asks whether

$$T_{f_1} \cdots T_{f_N} = 0 \tag{1.1}$$

for bounded symbols forces some $f_j = 0$ almost everywhere [1, Problem 1.1]. The source proves this when all but possibly one symbol are radial on the unit ball, allowing the exceptional L^2 symbol to occur between radial factors.

only if either f or g is holomorphic. From this it is readily deduced that if $f, g \in L^\infty$ such that $T_g T_f = 0$ then one of the symbols must be the zero function. A more general question concerning products of several Toeplitz operators (on H^2 or A^2) is the so-called “zero-product” problem.

Problem 1.1. *Suppose $f_1, \dots, f_N$ are functions in L^∞ such that $T_{f_1} \cdots T_{f_N}$ is the zero operator. Does it follow that one of the functions f_j 's must be the zero function?*

For Toeplitz operators on H^2 , the affirmative answer was proved by K.Y. Guo [6] for $N = 5$ and by C. Gu [5] for $N = 6$. Problem [1.1] for general N remains open. For Toeplitz operators on the Bergman space of the unit disk, Ahern and Čučković [1] answered Problem [1.1] affirmatively for the case $N = 2$ with an additional assumption that both symbols are bounded harmonic functions. In fact they studied a type of Brown-Halmos Theorem for Toeplitz operators on the Bergman space. They showed that if f and g are bounded harmonic functions and h is a bounded C^2 function whose invariant Laplacian is also bounded then the equality $T_g T_f = T_h$ holds only in the trivial case, that is, when f or $\bar{g}$ is holomorphic. As a corollary they obtained their solution to Problem [1.1]. This result was generalized to Toeplitz operators on the Bergman space of the unit ball in $\mathbb{C}^n$ by B. Choe and K. Koo in [4] with an assumption about the continuity of the symbols on an open subset of the boundary. They were also able to show that if $f_1, \dots, f_{n+3}$ (here $N = n + 3$) are bounded harmonic functions that have Lipschitz continuous extensions to the whole boundary of the unit ball then

Figure 1: Problem 1.1 and its open general Bergman case [1, p. 2].

The closest later results are precise. Dong and Zhou prove the two-factor case when one symbol is arbitrary and the other separately quasihomogeneous [2]. Cuckovic, Huo, and Sahutoglu prove the zero-product theorem for finite sums of quasihomogeneous symbols on bounded Reinhardt domains; their Corollary 1 permits one arbitrary symbol at the *left endpoint* and single-character factors to

its right [3]. The current March 2026 revision still identifies interacting arbitrary-symbol series as the obstruction. It does not cover an arbitrary factor with nontrivial quasihomogeneous strings on both sides.

quasi-homogeneous symbols.

In this paper we study zero products of finitely many Toeplitz operators acting on the Bergman space of bounded Reinhardt domains in $\mathbb{C}^n$, with the assumption that the symbols are finite sums of quasi-homogeneous functions. Our results point again in the direction of the trivial solution. But as opposed to the unit ball and the polydisk, working on more general Reinhardt domains brings new technical difficulties mainly stemming from the lack of explicit formulae for the Bergman kernels on general Reinhardt domains. Furthermore, additional difficulties come from working with finitely many products of Toeplitz operators and choosing the symbols that are finite sums of quasi-homogeneous functions. We hope that our techniques will bring more light to the study of algebraic properties of Toeplitz operators on more general domains in $\mathbb{C}^n$, beyond the unit ball and polydisk.

Our main result is the following theorem.

Theorem 1. *Let Ω be a bounded Reinhardt domain in $\mathbb{C}^n$ and $\phi_1, \dots, \phi_m$ be finite sums of bounded quasi-homogeneous functions. Assume that $T_{\phi_m} \cdots T_{\phi_1} = 0$ on $A^2(\Omega)$. Then $\phi_j = 0$ for some j .*

In particular, our result shows that, the type of examples in [BL11] and [ÇZ16] cannot happen on bounded domains. In addition, we obtain the following corollary.

Corollary 1. *Let Ω be a bounded Reinhardt domain in $\mathbb{C}^n$ and $\phi_1, \dots, \phi_{m-1}$ be bounded quasi-homogeneous symbols on Ω and let $\phi_m \in L^\infty(\Omega)$. Assume that $T_{\phi_m} \cdots T_{\phi_1} = 0$ on $A^2(\Omega)$. Then $\phi_j = 0$ for some j .*

In case there are two or more L^∞ symbols, the proof of Corollary 1 would involve multiple interacting series. Because of this difficulty, the zero product problem for Toeplitz operators is still open even on the unit disc for multiple L^∞ symbols. We believe new ideas are needed in that case.

— —

Figure 2: The closest later theorem, endpoint-arbitrary corollary, and stated obstruction in the March 2026 revision of [3, p. 2].

2 Idea of the proof

A separately quasihomogeneous Toeplitz operator has exactly one angular character, so it sends each sufficiently large monomial to one shifted monomial. Its scalar weight is a value of a multivariable Mellin transform. Consequently, a matrix coefficient of a product with one arbitrary middle symbol is a *scalar product* of a left Mellin factor, one fixed angular Fourier–Mellin transform of the arbitrary symbol, and a right Mellin factor.

Move the input and output multi-indices together along the positive integer lattice. Their angular difference stays fixed, and the scalar product vanishes at every lattice point. That lattice is a uniqueness set for bounded holomorphic functions on a product of right half-planes. The two outer factors are nonzero holomorphic functions, so the fixed-frequency transform of the arbitrary symbol must vanish. Repeating for every angular frequency recovers the entire symbol.

The single-character assumption is essential to this proof. Finite angular sums on both sides create collisions between different Fourier coefficients of the middle symbol; two arbitrary symbols create an infinite convolution.

3 The theorem

A bounded measurable a on a Reinhardt domain is *separately quasihomogeneous of degree* $p \in \mathbb{Z}^n$ if, away from coordinate hyperplanes,

$$a(r_1 e^{i\theta_1}, \dots, r_n e^{i\theta_n}) = A(r_1, \dots, r_n) e^{ip \cdot \theta} \quad (3.1)$$

for a bounded measurable radial amplitude A .

Theorem 1. *Let $\Omega \subset \mathbb{C}^n$ be a bounded Reinhardt domain. Let $a_1, \dots, a_r, b_1, \dots, b_s$ be bounded separately quasihomogeneous symbols and let $f \in L^2(\Omega)$. If*

$$T_{a_1} \cdots T_{a_r} T_f T_{b_1} \cdots T_{b_s} = 0 \quad (3.2)$$

on analytic polynomials, then at least one of $a_1, \dots, a_r, f, b_1, \dots, b_s$ is zero almost everywhere. Empty left or right strings are allowed.

Thus the exceptional arbitrary symbol may occur at any position. On the unit ball, degree-zero separately radial amplitudes already strictly contain the fully radial class used in the source; arbitrary degrees give weighted shifts rather than diagonal operators.

4 Mellin and weighted-shift lemmas

Write $t_j = |z_j|^2$ and let $E \subset \mathbb{R}_+^n$ be the radial image of Ω . A coordinate dilation lets us assume $E \subset [0, 1]^n$ without changing the argument.

Lemma 2 (integer-lattice uniqueness). *If F is bounded and holomorphic on $\mathbb{H}_+^n$ and $F(k) = 0$ for every $k \in \mathbb{N}^n$ with all $k_j \geq 1$, then $F \equiv 0$.*

Proof. For one variable, a nonzero bounded holomorphic function on $\mathbb{H}_+$ cannot vanish at all positive integers: after mapping $\mathbb{H}_+$ to the disk, their Blaschke sum diverges. For several variables, fix the last $n - 1$ variables at positive integers and apply the one-variable result in the first variable. Then fix an arbitrary first variable and positive integers in the remaining variables, apply the result in the second variable, and iterate. $\square$

Lemma 3 (one-character shift). *Let a have degree $p \in \mathbb{Z}^n$ as in (3.1). For every sufficiently large $\alpha \in \mathbb{N}^n$,*

$$T_a z^\alpha = c_a(\alpha) z^{\alpha+p}, \quad (4.1)$$

where a positive monomial-norm factor times $c_a(\alpha)$ equals

$$\mathcal{M}_a(\alpha) := C_\Omega \int_E A(\sqrt{t}) t^{\alpha+p/2} dt. \quad (4.2)$$

After translating α by a sufficiently large fixed integer vector, $\mathcal{M}_a$ extends to a bounded holomorphic function on $\mathbb{H}_+^n$. If $a \neq 0$ almost everywhere, that extension is not identically zero.

Proof. Angular orthogonality shows that $P(az^\alpha)$ has only torus character $\alpha + p$; for large α this character is the monomial $z^{\alpha+p}$. Polar integration gives (4.2). Choose a fixed shift so every power multiplying A is nonnegative. The shifted integrand is in $L^1(E)$, and $|t^u| \leq 1$ for $u \in \mathbb{H}_+^n$, proving bounded holomorphy by dominated convergence on compact sub-half-planes.

If the extension were zero, all polynomial moments of the shifted integrand would vanish. Extend it by zero from E to $[0, 1]^n$. Stone–Weierstrass then makes it zero almost everywhere. The monomial weight is positive off coordinate hyperplanes, so $A = 0$ almost everywhere, a contradiction. $\square$

5 Proof of Theorem 1

Assume every a_i and b_j is nonzero. We prove $f = 0$. Iterating Lemma 3, there are total degrees $p, q \in \mathbb{Z}^n$ such that, for sufficiently large $\beta, \alpha \in \mathbb{N}^n$,

$$T_{a_1} \cdots T_{a_r} z^\beta = L_0(\beta) z^{\beta+p}, \quad (5.1)$$

$$T_{b_1} \cdots T_{b_s} z^\alpha = R_0(\alpha) z^{\alpha+q}. \quad (5.2)$$

After removing positive monomial-norm factors, L_0 and R_0 have the same integer zeros as finite products L and R of fixed translates of the Mellin transforms (4.2). Each factor is nonzero by Lemma 3; hence L and R are nonzero holomorphic functions.

Take the coefficient of $z^{\beta+p}$ in (3.2) applied to z^α . After dividing only the positive norm factors, we obtain

$$L(\beta)R(\alpha) \int_{\Omega} f(z) z^{\alpha+q} \overline{z^\beta} dV(z) = 0 \quad (5.3)$$

for all sufficiently large integer α, β .

Fix an arbitrary $d \in \mathbb{Z}^n$. Choose large $\alpha_0, \beta_0 \in \mathbb{N}^n$ so that every intermediate weighted-shift index is nonnegative and

$$\alpha_0 + q - \beta_0 = d. \quad (5.4)$$

For $u \in \mathbb{H}_+^n$, set

$$B_d(u) = L(\beta_0 + u)R(\alpha_0 + u), \quad (5.5)$$

$$G_d(u) = \int_{\Omega} f(z) z^{\alpha_0+q} \overline{z^{\beta_0}} \prod_{j=1}^n |z_j|^{2u_j} dV(z). \quad (5.6)$$

All fixed powers can be enlarged when choosing α_0, β_0 , so polar coordinates express G_d as the Mellin transform of an $L^1(E)$ function. Thus G_d is bounded holomorphic on $\mathbb{H}_+^n$. The same holds for B_d , and $B_d \not\equiv 0$.

Putting $\alpha = \alpha_0 + k$ and $\beta = \beta_0 + k$ in (5.3) gives

$$B_d(k)G_d(k) = 0 \quad (k \in \mathbb{N}^n). \quad (5.7)$$

Lemma 2 implies $B_d G_d \equiv 0$. The ring of holomorphic functions on the connected domain $\mathbb{H}_+^n$ has no zero divisors, and $B_d \not\equiv 0$, so

$$G_d \equiv 0. \quad (5.8)$$

For integer u , equation (5.6) gives every polynomial moment of the $(-d)$ -th torus Fourier coefficient of f , multiplied by the fixed positive monomial weight from (5.6). Moment uniqueness on $[0, 1]^n$ makes that coefficient zero almost everywhere on E . Since $d \in \mathbb{Z}^n$ was arbitrary, all torus Fourier coefficients of f vanish. Fubini's theorem and Fourier uniqueness in $L^1(\mathbb{T}^n)$ give $f = 0$ almost everywhere, completing the proof.

6 Consequences and eight upgrade attempts

Corollary 4. *On the unit ball, Le's theorem remains true when every radial factor is replaced by an arbitrary bounded separately quasihomogeneous factor, of any degree, and the exceptional L^2 symbol is placed anywhere in the product. The same statement holds on every bounded Reinhardt domain.*

Eight focused routes were audited. A current-status search confirmed that the general Bergman problem is still open and separated the 2011 two-factor and 2020 endpoint results from Theorem 1. Replacing full radial symmetry by coordinate-torus invariance succeeds through multivariable Mellin uniqueness. Allowing nonzero torus degrees succeeds because the operators remain one-character weighted shifts. Moving the arbitrary factor into the interior succeeds by the two-sided matrix coefficient (5.3). Passing from the ball to bounded Reinhardt domains succeeds because only torus orthogonality and moment uniqueness are used.

Three stronger routes fail at a precise point. Finite character sums on both sides allow different outer degrees and different Fourier coefficients of f to collide in one output character, replacing (5.7) by an interacting sum. Two arbitrary symbols produce an infinite convolution of two Fourier series. Finally, direct cancellation of the outer strings is unavailable: nonzero quasihomogeneous weighted shifts can have sparse zero weights and need not be injective or have dense range. No verified argument removes any of these obstructions.

Theorem 1 is therefore a full result for its stated class and a substantial partial answer to Problem 1.1, not a solution of the unrestricted problem.

7 Verification and novelty bounds

The proof was checked independently at the four sensitive points: all intermediate exponents can be made nonnegative; positive norm factors are discarded only on the integer lattice; every shifted outer Mellin transform is nonzero; and fixed angular difference turns the arbitrary matrix moment into one Mellin transform. No computational assertion is used.

The run indexes were searched for arXiv:0712.0167, “zero-product Toeplitz”, “separately quasihomogeneous”, “arbitrary symbol”, and “Reinhardt”. Bounded primary-source searches through 13 August 2026 checked Le’s source, Dong–Zhou’s 2011 paper, arXiv:2009.01951 in its March 2026 revision, and recent papers that still describe the arbitrary-symbol Bergman problem as open. They found endpoint placement and two-factor results, but no theorem with quasihomogeneous strings on both sides of an arbitrary factor. Novelty confidence is moderate because the literature search was bounded.

Promote for expert review. The reviewer should focus on the cumulative degree convention in (5.1)–(5.3) and the standard Laurent-monomial decomposition on a general Reinhardt domain.

References

- [1] T. Le, *The zero-product problem for Toeplitz operators with radial symbols*, arXiv:0712.0167 (2007).
- [2] X.-T. Dong and Z.-H. Zhou, *Algebraic properties of Toeplitz operators with separately quasihomogeneous symbols on the Bergman space of the unit ball*, J. Operator Theory 66 (2011), 193–207.
- [3] Z. Cuckovic, Z. Huo, and S. Sahutoglu, *Zero products of Toeplitz operators on Reinhardt domains*, arXiv:2009.01951, revision of 22 March 2026.